

4. Real functions - Limit

Calculus

Finite limit at a finite point

Let $D \subseteq \mathbb{R}$, $f : D \rightarrow \mathbb{R}$, and $p \in \mathbb{R}$ be an accumulation point of D .

• **Definition (Finite one-sided limit).** We say that the *right hand side* (resp. *left hand side*) *limit of f at p* exists if there exists $A \in \mathbb{R}$ such that, for all $\varepsilon > 0$, one can find $r > 0$ such that $|f(x) - A| < \varepsilon$, whenever $x \in]p, p + r[\cap D$ (resp. $x \in]p - r, p[\cap D$). The number A is called the right hand side (resp. left hand side) limit of f at p ; notation:

$$\lim_{x \rightarrow p^+} f(x) := A \quad \left(\text{resp.} \quad \lim_{x \rightarrow p^-} f(x) := A \right).$$

• **Definition (Limit).** We say that the limit of f at the point p exists if there exists $A \in \mathbb{R}$ such that, for all $\varepsilon > 0$ one can find $r > 0$ such that $|f(x) - A| < \varepsilon$ if $|x - p| < r$. Then the number A is called the limit of f at the point p and denoted by

$$\lim_{x \rightarrow p} f(x) := A.$$

• **Theorem.** The limit of f at p exists if and only if its one sided limits exist and coincide at the point p .

Limit and continuity

- **Theorem (Limit and continuity).** A function is continuous at a given accumulation point of its domain if and only if its limit exists at the point and equals to the substitution value at the point.

Proof. Homework...

Infinite limit at a finite point

Let $D \subseteq \mathbb{R}$, $f : D \rightarrow \mathbb{R}$, and $p \in \mathbb{R}$ be an accumulation point of D .

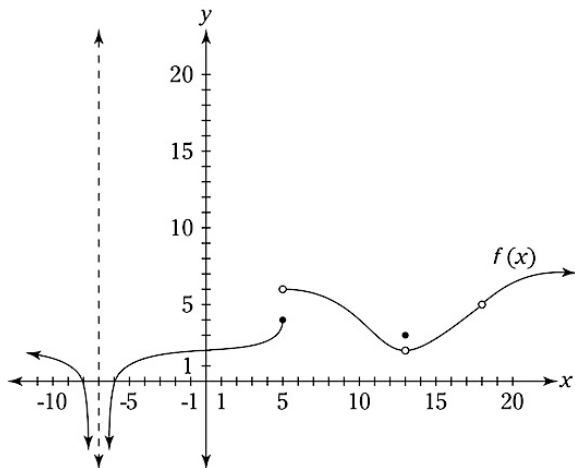
• **Definition (Infinite one-sided limit).** We say that the right hand side limit (resp. the left hand side limit) of f at the point p is $+\infty$ if, for all $M \in \mathbb{R}$, one can find $r > 0$ such that $f(x) > M$, whenever $x \in]p, p + r[$ (resp. $x \in]p - r, p[$).

$$\lim_{x \rightarrow p^+} f(x) := +\infty \quad \left(\text{resp. } \lim_{x \rightarrow p^-} f(x) := +\infty \right).$$

Similarly, we say that the right hand side limit (resp. the left hand side limit) of f at the point p is $-\infty$ if, for all $M \in \mathbb{R}$, one can find $r > 0$ such that $f(x) < M$, whenever $x \in]p, p + r[$ (resp. $x \in]p - r, p[$).

$$\lim_{x \rightarrow p^+} f(x) := -\infty \quad \left(\text{resp. } \lim_{x \rightarrow p^-} f(x) := -\infty \right).$$

Limits taken finite points

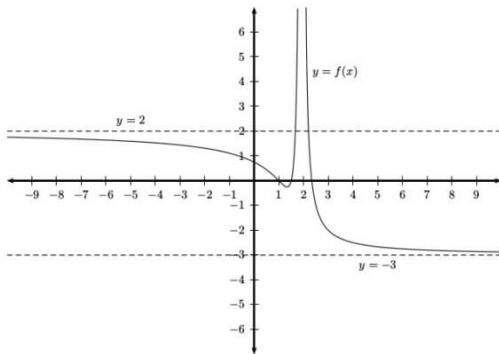


Finite limit at infinity

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$.

• **Definition (Finite limit at infinity).** We say that *the limit of f exists at $+\infty$ (resp. at $-\infty$)* if $+\infty$ (resp. $-\infty$) is an accumulation point of D and there exists $A \in \mathbb{R}$ such that, for all $\varepsilon > 0$, one can find $\xi \in D$ such that $|f(x) - A| < \varepsilon$, whenever $x \in]\xi, +\infty[\cap D$ (resp. $x \in]-\infty, \xi[\cap D$). Notation:

$$\lim_{x \rightarrow +\infty} f(x) := A \quad \left(\text{resp. } \lim_{x \rightarrow -\infty} f(x) := A \right).$$



Infinite limit at infinity

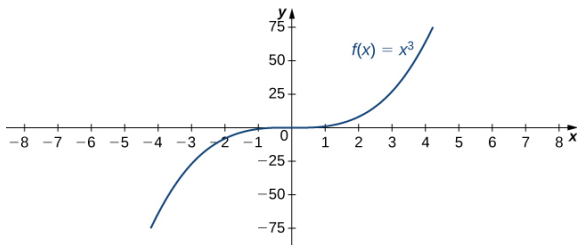
Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$.

• **Definition (Infinite limit at infinity).** We say that *the limit of f at $+\infty$ (resp. at $-\infty$) is $+\infty$* if $+\infty$ (resp. $-\infty$) is an accumulation point of D and, for all $M \in \mathbb{R}$, there exists $\xi \in D$ such that $f(x) > M$, whenever $x \in]\xi, +\infty[\cap D$ (resp. $x \in]-\infty, \xi[\cap D$).
Notation:

$$\lim_{x \rightarrow +\infty} f(x) := +\infty \quad \left(\text{resp. } \lim_{x \rightarrow -\infty} f(x) := +\infty \right).$$

Similarly, say that *the limit of f at $+\infty$ (resp. at $-\infty$) is $-\infty$* if $+\infty$ (resp. $-\infty$) is an accumulation point of D and, for all $M \in \mathbb{R}$, there exists $\xi \in D$ such that $f(x) < M$, whenever $x \in]\xi, +\infty[\cap D$ (resp. $x \in]-\infty, \xi[\cap D$). Notation:

$$\lim_{x \rightarrow +\infty} f(x) := -\infty \quad \left(\text{resp. } \lim_{x \rightarrow -\infty} f(x) := -\infty \right).$$



Limit of functions and sequences

• **Theorem.** Let $D \subseteq \mathbb{R}$ be a non-empty set, $p \in \mathbb{R} \cup \{-\infty, +\infty\}$ be an accumulation point of D and $f : D \rightarrow \mathbb{R}$. Let further $\Lambda \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then we have

$$\lim_{x \rightarrow p} f(x) = \Lambda$$

if and only if for all real sequences $(x_n) \subseteq D$ converging to $p \in \mathbb{R}$ or diverging to $p \in \{-\infty, +\infty\}$ we have $f(x_n) \rightarrow \Lambda$ as $n \rightarrow \infty$.

Limits and operations

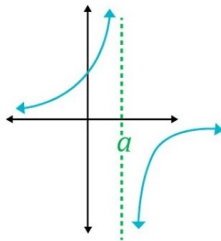
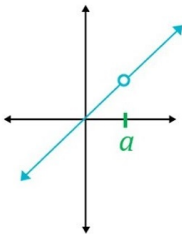
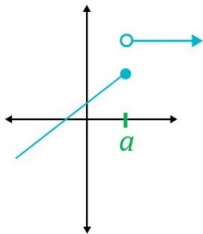
• **Theorem (Limit and operations).** Let $D \subseteq \mathbb{R}$ be a non-empty set, $p \in \mathbb{R} \cup \{-\infty, +\infty\}$ be an accumulation point of D , and $f, g : D \rightarrow \mathbb{R}$. Assume that the limit A and B of f and g exist at the point p , respectively. Then the following statements hold.

- (1) The limit of $f + g$ at p exists and equals to $A + B$.
- (2) The limit of $f \cdot g$ at p exists and equals to $A \cdot B$.
- (3) If $Bg(x) \neq 0$ for all $x \in D$, then the limit of $\frac{f}{g}$ at p exists and equals to $\frac{A}{B}$.

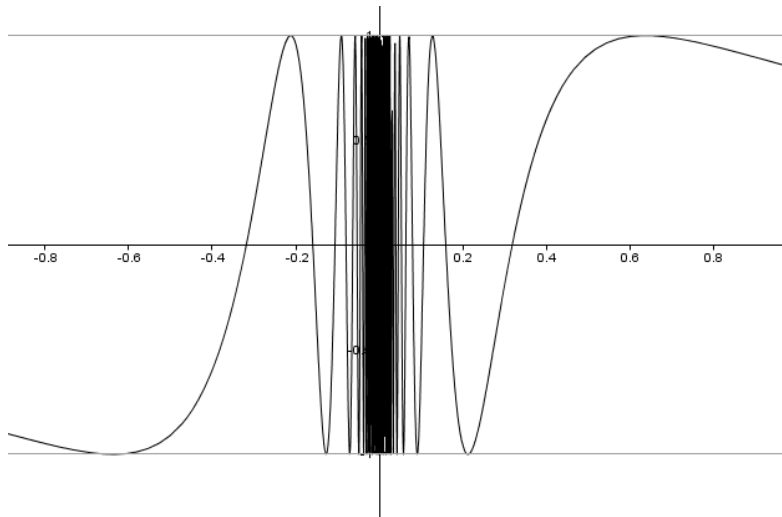
Classification of discontinuities

● **Definition.** Let $D \subseteq \mathbb{R}$ be a non-empty open set, $f : D \rightarrow \mathbb{R}$, and $p \in D$ be a discontinuity of f .

- ★ We say that p is a discontinuity of the first kind if the one-sided limits of f at p exist. If the one-sided limits are different, then p is called a *jump discontinuity*, otherwise we say that it is a *removable discontinuity*.
- ★ If p is not a discontinuity of the first kind, then it is called a discontinuity of the second kind.



Discontinuity of the second kind (further example)



Continuity of rational fractions

- **Continuity of a rational fraction.** A real number is a discontinuity of a rational fraction if and only if it is a zero of its denominator.
- Investigation of a rational fraction for continuity:
 - ★ *To find the discontinuities, calculate the zeros of the denominator.*
 - ★ Compute the one-sided limits at each discontinuity and the limits at $\pm\infty$. The calculation of the one-sided limits is facilitated by knowing the type of discontinuities: for a rational fraction, this can either be a removable or one in which the one-sided limits exist in the extended sense.
 - ★ The type of discontinuities can be determined by calculating the zeros of the numerator.

Let p be a discontinuity of f and suppose that it is a root of multiplicity $n_p \in \mathbb{N}$ and $m_p \in \mathbb{N} \cup \{0\}$ of the denominator and the numerator, respectively.

- If $m_p < n_p$ then p is a discontinuity of second kind. If $n_p - m_p$ is even, then the one-sided limits existing in the extended sense are the same, otherwise they are different.
- If $n_p \leq m_p$ then p is removable.